9 January 1991
SUPERSEDING
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MILITARY SPECIFICATION

INDICATOR, OIL QUANTITY, CONTINUOUS TRANSMISSION, NUCLEONIC

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for a nucleonic, continuous transistorized, aircraft oil quantity gaging system.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-L-20	Lacquer, Camouflage
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-W-5086	Wire, Electric, Polyvinyl Chloride Insulated, Copper or Copper Alloy
MIL-E-5400	Electronic Equipment, Aerospace, General Specification For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, Nato Code Number O-148

beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6680

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MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges.
MIL-W-22759	Wire, Electric, Fluoropolymer Insulated, Copper or Copper Alloy
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-C-25516/17	Connector, Electrical, Miniature Coaxial Environmental Resistant Type, Coaxial Hermetically Sealed
MIL-C-25516/20	Connector, Electrical, Miniature, Coaxial, Environment Resistant Type, Coaxial, Plug, Series 1, Class G, Type 1
MIL-L-27160	Lighting, instrument, Integral, White, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	Electromagnetic Intereference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-701	Lists of Standards Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-756	Reliability Modeling and Prediction
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-859	Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods.
MIL-STD-1523	Age Controls of Age-Sensitive Electromeric Material.
MS27034	Connector, Receptable, Electrical, Pin Insert, Cylindrical, Miniature, Hermetic, Solder Mount
* MS27572	Case, Instrument, Flange-mounted, Aircraft.
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS33639	Case, Instrument, Clamp-mounted, Aircraft.

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PUBLICATIONS

MILITARY

MIL-HDBK-217 Reliability Prediction of Electronic Equipment.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

U.S. Nuclear Regulatory Commission

Title 10, part 20, Standards for Protection Against Radiation

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402, or contact your Base Bioenvironmental Engineering Office.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Packaging, Commercial

*(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First Article - When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Components. The oil quantity gaging system shall consist of the following components.

a. Gaseous electromagnetic radiation source

b. Radiation detector

c. Master indicator consisting of the electronics necessary to produce a continuous indication of oil quantity

d. When required, a repeater indicator to produce a continuous indication of oil quantity when signals are received from the master indicator.

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e. Bridle assembly including brackets and hardware for mounting the gaseous electromagnetic radiation source and detector on the oil tanks.

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials used in the gaging system shall be in accordance with the requirements of MIL-E-5400. Materials that are not covered by applicable specifications or are not specifically described herein shall be of the best commercial quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. If used within a hermetically sealed inclosure, fungicidal treatment will not be required.

3.4.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the system except where magnetic materials are essential.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to atmospheric and climatic conditions likely to be encountered in storage or normal service.

3.4.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.5 Synthetic materials. For parts manufactured of synthetic materials, manufacturers shall maintain quality control of subsequent batches to provide for uniformity.

3.4.5.1 Age controls. Age controls for synthetic rubber parts shall comply with the requirements of MIL-STD-1523.

* 3.4.5.2 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.4.6 Magnesium and zinc. Magnesium and zinc shall not be used where they may be exposed to fuel, oil, or their vapors.

3.4.7 Toxic and corrosive fumes. The materials, as installed and under the service conditions specified, shall not liberate gases or fumes which will result in corrosion that will cause malfunction of equipment or discoloration of dials or indicia. Toxic gases or fumes that are detrimental to performance of the aircraft or health of personnel shall not be liberated under the service conditions specified herein.

3.4.8 Protective treatment. When materials are used in the construction of the gaging system that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of

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any protective coating that will chip, crack, or scale with age or extreme climatic environmental conditions shall be avoided.

3.5 Design and construction. The system shall provide continuous gaging of the useable oil in an engine oil tank through the entire range of full through empty. Full means the maximum level to which the tank is normally serviced and empty means that all useable oil has been drained from the tank and that only the unuseable oil, if any, remains in the sump area of the tank. The system shall be designed to measure the attenuation of the electromagnetic rays through the oil and the tank, convert the measurement to a signal suitable for the operation of an indicator, and provide a continuous visual display of the quantity of oil in the tank. The sources and detector shall be entirely external to the tank.

3.5.1 Input power. The gaging system shall perform in accordance with the requirements specified herein when supplied with 115 Volt (V), 400-Hertz (Hz) singlephase power in accordance with MIL-STD-704 except the voltage variation shall be 115 ± 10V and frequency variation shall be 400 ± 40 Hz. The total power consumption shall not exceed 8 watts per engine installation.

3.5.2 Screw threads. Unless otherwise specified, screw threads shall be in accordance with MIL-S-7742.

3.5.3 Locking devices. All internal threaded parts shall be positively locked in accordance with the requirements of MIL-E-5400.

3.5.4 Transistors and diodes. Transistors and diodes used in the system shall be selected in accordance with MIL-STD-701. Electron vacuum tubes shall not be used.

3.5.5 Reliability and longevity.

3.5.5.1 Indicators. The master and repeater indicators shall have a minimum acceptable mean-time-between failures (MTBF) of 667 hours (specified MTBF of 1,000 hours) at a confidence factor of 90 percent and a longevity of 3,000 hours.

3.5.5.2 Detector. The detector shall have a minimum acceptable MTBF of 2,000 hours (specified MTBF of 3,000 hours) at a confidence factor of 90 percent and a longevity of 3,000 hours.

3.5.5.3 Reliability analysis and test program. The reliability performance for the indicators and detectors shall be substantiated through a reliability design analysis, as specified in 3.5.5.3.1 and 3.5.5.3.2 and testing as specified in 4.6.12.

3.5.5.3.1 Failure effect analysis. The failure effect analysis shall be a qualitative analysis evaluating the effects of various modes of system failure as it affects performance and safety. Each unit in the system shall be considered individually and its failure mode determined. A failure of the unit shall then be assumed in each of the modes, and the remainder of the system examined to determine the overall effect of the failure. This process shall be repeated for every failure mode of each unit in the system.

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3.5.5.3.2 Numerical analysis. The degree of reliability of all components shall be determined by a pretest numerical analysis and shall be given in mean-time-between-failures. All analysis shall be in strict accordance with the methods prescribed in MIL-STD-756 and the calculation based on data obtained from MIL-HDBK-217. Where data is not available from this source, the calculations shall be based on past experience or industry standard failure rate data.

3.5.6 Maintainability program. The contractor shall establish a maintainability program in accordance with MIL-STD-470.

3.5.6.1 Maintainability. The following maintainability requirements shall apply:

a. Minimization of complexity of maintenance tasks by maximum use of simple design which includes optimum interchangeability and use of standard parts and equipment.

b. Require minimum number and types of tools and test equipment (special and standard) needed to perform maintenance.

c. Require minimum number and types of spare parts and assemblies.

3.6 Performance. The gaging system shall operate in accordance with the requirements of this specification under any combination of the following conditions:

a. Temperature

(1)	Indicators	-62° to +93°C (-80° to +199°F) (nonoperating) -54° to +71°C (-65° to 160°F) (operating)
(2)	Detectors	-62° to +204°C (-80° to 406°F) (nonoperating) -40° to +204°C (-40° to 406°F) (operating)

b. Altitudes - Pressures equivalent to altitudes from -1,000 to +100,000 feet (ft)(305 to 30480 meter(M)), as defined in MIL-STD-859.

c. Humidity - Relative humidity up to 95 percent

d. Salt fog - Exposure to moist salt laden atmosphere

e. Vibration - As specified in section 4

f. Fungus - Fungus growth, as encountered in tropical climates.

g. Thermal shock - Thermal shock between 5° and +85°C (41° and 185°F).

3.6.1 Accuracy. For all tank attitudes between +50° and -20° pitch and ±15° roll, the allowable error of the gaging system over the entire range of oil quantities from full through empty shall be as specified in table I. For tank attitudes intermediate to those listed in table I, the next lowest specified value for attitude shall be used. The following are examples:

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TABLE I. Tolerance in Fraction of Full-Scale Indication for Pitch-and-Roll Attitudes

Pitch \ Roll	0°	±5°	±10°	±15°
-20°	2/32	2/32	2/32	3/32
-10°	1/32	1/32	2/32	3/32
0°	1/32	1/32	2/32	3/32
+10°	1/32	1/32	2/32	3/32
+20°	1/32	2/32	2/32	3/32
+30°	2/32	2/32	2/32	3/32
+40°	2/32	2/32	2/32	3/32
+50°	3/32	3/32	3/32	3/32

Note: Accuracy requirements apply to readings taken from both the master and repeater indicators.

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Tank Attitude	Table I Value	Tolerance
+25° pitch, 0° roll	+20° pitch, 0° roll	±1/32
-19° pitch, 9° roll	-10° pitch, 5° roll	±1/32
+17° pitch, 12° roll	+10° pitch, 10° roll	±2/32

3.6.2 Electromagnetic interference. The system shall meet the electromagnetic interference and susceptibility requirements of MIL-STD-461 for class ID equipment.

3.7 Interchangeability. Gaging system components shall meet the interchangeability requirements of MIL-I-8500.

3.8 Details of components

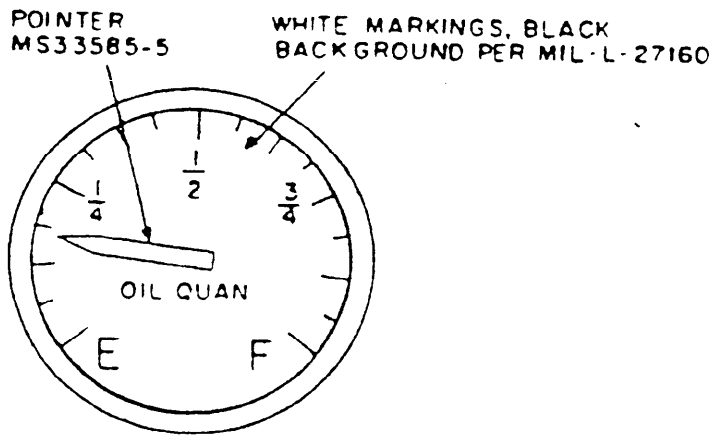
3.8.1 Detector. The detector shall be rigidly mounted to the tank and suitably protected to prevent damage during installation and use. No holes through the tank or rework of the tank will be permitted for mounting the detector unit. In addition to withstanding the aircraft environmental conditions, the detector shall withstand the bumps, jars, et cetera, incident to handling and installation (including misassembly and overtorquing of the bridle assembly). A standard Military-type connector shall be mounted, without the use of cable, directly to the detector assembly.

3.8.1.1 Dielectric strength. The detector shall withstand, without breakdown or other damage, a voltage equal to 1.5 times the maximum operating voltage applied between the center conductor and the case.

3.8.2 Radiation source. The radiation source shall be installed on the oil tank in its proper location without the use of special tools or equipment and without any modification or rework of the oil tank. The radioisotope shall be Krypton (Kr) 85 gas. The maximum activity of Kr 85 per single tube shall not exceed 500 millicurries (mc) and the maximum activity per engine installation shall not exceed 800 mc. The source strength at the time of delivery shall not be below the design source strength by more than 2 percent. The maximum radiation exposure rate at a distance of 1 ft (0.3048 m) from the tank shall not exceed 4 milliroentgens per hour (mr/hr) when tested as specified in section 4. If necessary, the source shall be shielded with a suitable material to limit the radiation exposure rate to the maximum specified herein. The method of containment shall preclude release of the gas under normal handling and operating conditions and under emergency conditions not involving exposure to major stress. The gas shall be contained without leakage when exposed to the shock and fire tests.

3.8.3 Master indicator. The master indicator shall be capable of receiving the electrical output of the detector and driving a dial pointer which continuously indicates the amount of oil present in the tank. The dial face of the indicator shall be in accordance with figure 1. The display shall have an indicated scale from full to empty of 235° with a minimum of 10° overtravel at the full end and a minimum of 5° overtravel at the empty end. Stops shall be incorporated to prevent pointer travel beyond the 250° total range. The master indicator shall also furnish signals for the operation of a repeater indicator. These signals shall be provided independent of any failure in the mechanism to move the dial pointer.

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MARKINGS	HEIGHT OR LENGTH ±0.010	WIDTH OF LINE OR GRAD ±.0005
NUMERALS	0.125	0.025
OIL QUAN	0.140	0.020
MAJOR GRAD	0.188	0.031
INTERMEDIATE GRAD	0.160	0.020
MINOR GRAD	0.125	0.015
E AND F	0.200	0.031

FIGURE 1. Dial Face

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3.8.3.1 Warmup time. The indicator shall operate as specified herein within 1 minute after power input under any combination of environmental extremes specified herein.

3.8.3.2 Low level warning provisions. Provisions shall be made in the master indicator for switching a low level warning light. Provisions shall be made to close the light circuit at a predetermined level upon descending oil level and to open this circuit at some higher predetermined level on ascending level. Switch contacts shall be capable of making and breaking at least 0.3 amp at 28V direct current or alternating current (dc or ac). The low oil level warning provisions shall be used only to indicate low oil quantities. These provisions shall not be designed to indicate system/component failures. The switching mechanism shall actuate independently of the mechanical operation of the pointer.

3.8.3.2.1 Circuit adjustment. The warning light actuator mechanism shall be adjustable from the rear surface of the indicator to provide closing of the circuit within ±1/32 of any point between 1/8 and 3/4 of full indication. Opening of the circuit shall be at 1/8 ± 1/32 of full scale above the adjusted closing position of the switch. The adjustment shall be permanently marked as to its function.

3.8.3.3 Calibration. Controls shall be provided on the rear of the indicator case to permit calibration of the indicator. These controls shall have sufficient range to allow the continued calibration and use of the indicator within the requirements of 3.6.1 for a period of 10 years when used with sources provided in accordance with 3.8.2 which have been used with the indicator for 10 years. Each control shall be permanently marked as to its function.

3.8.3.4 Dielectric strength. Insulation for electrical circuitry shall be sufficient to withstand a potential of 500V ac root-mean-square (rms) between the circuitry and the case for those circuits terminating at the power connector. The coaxial connectors and cabling used between the indicator and detector shall be capable of withstanding normal operating voltages in an environment equivalent to a 100,000 H (3048M) altitude without insulation breakdown.

* 3.8.3.5 Sealing. The indicator shall be hermetically sealed. The filling medium in the case shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The filling medium shall be at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint - 65°C (-85°F)) at the filling pressure. The absolute pressure of the filling medium shall be approximately 1 atmosphere. Leakage shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere. Alternate sealing methods may not be used without prior approval of the Contracting Officer.

3.8.3.6 Dial visibility. At least one-fourth of the shortest graduation shall be visible from any point within the frustum of a cone the side of which makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the instrument case. The dial face shall be flat.

3.8.3.7 Coverage and lighting wedge. The coverglass shall be suitable for hermetic sealing and shall be clear, flat, and free from flaws which would interfere with the normal reading of the instrument. The thickness of the coverglass shall be the minimum practicable and entirely suitable for the purpose intended. The distance between the front surface of the bezel and the

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outside surface of the coverglass shall not exceed 0.062 inch (1.58 milimeter (mm)). The distance between the inside surface of the coverglass and the dial shall not exceed 0.125 inch (3.18 mm). All reflecting glass surfaces shall be provided with a reflection-reducing coating that meets the requirements of MIL-C-14806.

3.8.3.8 Indicator lighting. The indicator shall incorporate white integral lighting in accordance with MIL-L-27160. Red integral lighting, if specified by the procuring activity in lieu of white lighting, shall be in accordance with MIL-L-25467.

3.8.3.9 Indicator response time. The indicator shall respond within 8.5 ±0.5 seconds to a one-half scale reading when subjected to a full-scale step-input change in oil level.

3.8.3.10 Indicator scale error. The scale error shall be less than ±1/32 of full scale when inputs equaling oil levels of 1/16, 1/4, 1/2, 3/4, and full are provided to the indicator.

3.8.3.11 Magnetic effect. The indicator, both operating and nonoperating, when positioned 5 ±0.5 inches (127 ±12.70 mm) from a magnetic compass within a field intensity of 0.17 to 0.19 oersted, shall not produce a compass deflection of more than 5°.

3.8.3.12 Pointer oscillation. The indicator pointer shall not oscillate with a peak to peak magnitude greater than 1/32 of full scale and with a frequency greater than 1 Hz.

3.8.4 Repeater indicator. The repeater indicator shall be capable of receiving the repeater signal from the master indicator and of duplicating the indication shown on the master indicator. The accuracy requirements of 3.6.1 also apply to the readings of the repeater indicator. The dial presentation shall be identical to that of the master indicator and warmup time, calibration, dielectric strength, sealing, dial visibility, coverglass and lighting wedge, lighting, response time, scale error, magnetic effect, thermal shock, and pointer oscillation requirements shall be the same as specified for the master indicator.

* 3.8.5 Indicator cases. The master and repeater cases shall be in accordance with either MS33639 for the 2-inch (50.8 mm) round case or MS27572 for the 2 inch case, except that the case shall be square, dimensions A and E do not apply, and dimension G is the maximum dimension of a side of the square case. The length of the master indicator, including the connector, shall not exceed 7 inches (177.80 mm). The length of the repeater indicator, including the connector, shall not exceed 4 inches (101.60 mm).

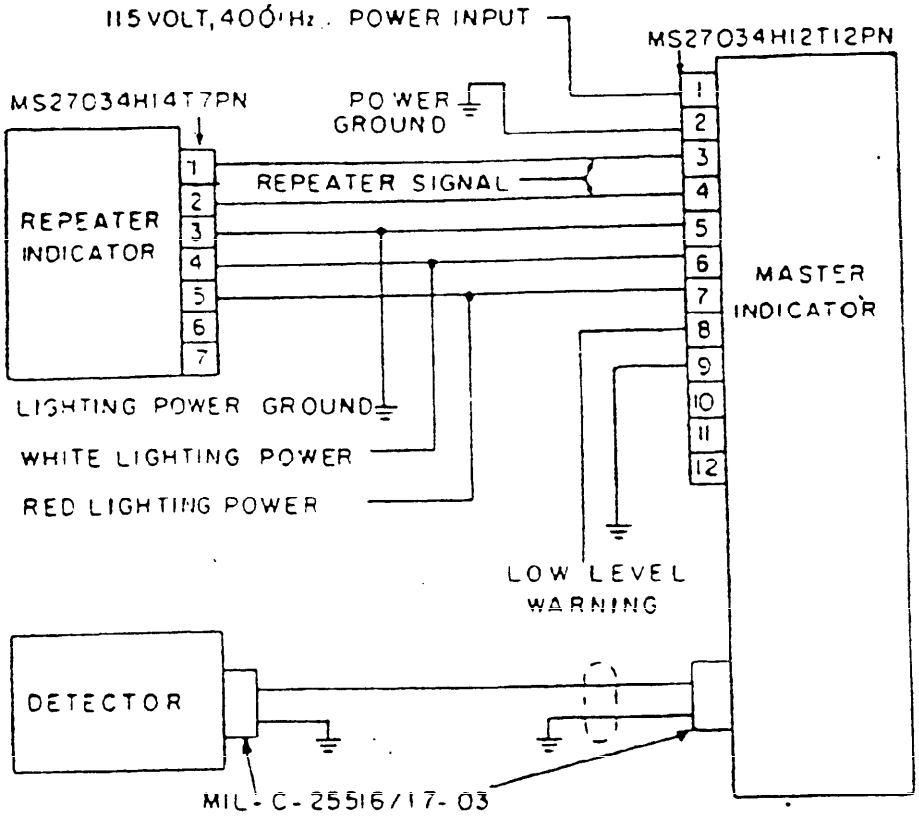
3.8.6 Connectors. The connectors shall be in accordance with MS27034 and MIL-C-25516 and shall be of the type specified on figure 2.

3.8.6.1 Electrical hookup. The components shall be capable of operation in accordance with the requirements specified herein when connected through an external hookup as shown on figure 2. The detector and master indicator shall perform equally well when the repeater indicator is not utilized.

* 3.8.7 Bridle assembly. The bridle assembly shall contain the source and detector on the oil tank. The design shall be submitted for approval and shall

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shall be at least 95 percent covered with a new solder coating. The remaining 5 percent of the lead surface



NOTES 1. Use extra pins if additional repeater signal lines are needed.

2. The lighting lines may also be provided individually

3. The Mil-C-25516/17-03 connector shall mate with a Mil-C-25516/20-03-07 connector.

FIGURE 2. External Hookup

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detectives in excess of those allowed in table III shall be ... 3.4.3 detectives

be so designed as to prevent chafing of the bridle, source, or detector on any portion of the engine, oil tank, or airframe during normal service and when removing the engine or oil tank (or both) from the airframe. The design shall be such as to prevent excessive stress on the source and detector which might cause them to fail during service use.

3.9 Cabling. The gaging system shall be designed to use cable of radio frequency, coaxial, dual coaxial, twin conductor and twin lead type between the indicator and detector. Installation of the cabling shall be such as to preclude failures due to excessive bending or chafing.

* 3.9.1 Wiring. Wiring between components, exclusive of cabling, shall be in accordance with MIL-W-5086 or MIL-W-22759.

3.10 Weight. The total weight per engine installation, exclusive of cabling, shall not exceed 5 pounds.

3.11 Finish. The gaging system components shall be finished in accordance with the requirements of MIL-E-5400. The lacquer for the indicator case, other than the connector and coverglass, shall conform to TT-L-20, color 37038 of FED-STD-595.

3.12 Identification of product. The gaging system components shall be marked for identification in accordance with MIL-STD-130. The source tube shall also be marked in accordance with the requirements of US Nuclear Regulatory Commision (NRC) regulation 10 NFR part 20. The source tube nameplate shall also include the NRC license number and serial number. A protective sleeve of clear heat-shrinkable plastic shall be installed over the source tube nameplate to prevent loss and to protect it from inadvertent damage or defacing.

3.13 Workmanship. The system shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.13.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the system, they shall be held or limited accordingly.

3.13.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.13.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.13.4 Cleaning. The system shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, or other foreign material after assembly. Burr and sharp edges as well as resin flash which might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified

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in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the gaging system shall be classified as follows:

* a. First Article Inspection (see 4.4)

b. Acceptance tests.

4.3 Test Conditions

4.3.1 Atmosphere conditions. Unless otherwise specified, all tests shall be made at an atmospheric pressure of 28 to 32 inches (71 to 81.12 Centimeter (cm)) Hg, at a temperature of 24° ±9°C (75° ±48°F), and at a relative humidity of 90 percent or less.

4.3.1.1. Temperature stabilization. All parts of the test item shall be completely stabilized at the specified temperature prior to conducting any tests.

4.3.2 Test power. Unless otherwise specified, direct-current voltage shall be maintained at 28 ±1V and alternating-current voltage shall be maintained at 115V ±2 percent 400 Hz ±2 percent.

4.3.3 Heat source. The heat source of the test facilities shall be so located that radiant heat will not fall directly on the equipment under test.

4.3.4 Measurements. All measurements shall be made with instruments on which the accuracy has been verified. These measurements shall allow for all correction factors.

4.3.5 Test oil. The test oil for all tests shall be in accordance with MIL-L-7808.

4.3.6 Cable length. A minimum of 50 feet (15.12 m) of cabling between the detector and master indicator and a minimum of 10 feet (3.05 m) of wiring between the master and repeater indicators shall be utilized in all tests.

4.3.7 Oil quantity. Unless otherwise specified, all tests shall be performed as system tests with oil tank quantities of 1/16, 1/4, 1/2, 3/4, and full at 0° pitch and 0° roll.

4.3.8 Indicator readings. The indicator readings (both master and repeater) rshall be the average visual quantity taken over a minimum 15-second period to the nearest 1/32 part of a full scale reading.

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4.3.9 Survey meter. The survey meter used to determine exposure rates shall be a Victoreen model number 440 or equivalent. The batteries shall be fresh and the meter shall have been calibrated with a Cesium 137 source within 15 days prior to the tests. The 3 or 10 mr/hr scales shall be used for all measurements. The beta shield shall be in place for all measurements..

4.3.10 Pretest performance record. Unless otherwise specified, the data taken during the calibration test shall constitute the pretest performance record required by MIL-STD-810.

4.3.11 Calibration. Unless otherwise specified, system calibration shall be accomplished only once for each test setup.

* 4.4 First Article Inspection. The first article tests shall consist of all tests specified under 4.6. A J79-11A engine oil tank, General Electric Part Number 107R111P2, or equivalent shall be used.

4.4.1 First Article Inspection Procedures. The contractor shall prepare the procedures used for conducting the first article inspections. The inspections shall include all the procedures specified in the specification, (Section 4). The inspection procedures shall be submitted to the procuring activity for review and approval. The procuring activity reserves the right to modify the inspections or require any additional inspections deemed necessary to determine compliance with the requirements of this specification and the contract. The first article inspection procedures shall be prepared to follow the requirements of this specification, paragraph by paragraph and from beginning to end. All visual check and operation examinations shall be included.

4.4.2 First Article Inspection Report. The report documenting the results of the first article inspections shall be prepared by the contractor. Two copies of the test report shall be submitted to the procuring activity for review. The first article inspection report shall be prepared in accordance with MIL-STD-831.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests

4.5.1 Individual tests. Each detector, master indicator, and repeater indicator (when required) shall be subjected to the following tests as described under 4.6, as applicable:

a. Examination of product

b. Detector burn-in

c. Dielectric strength

d. Indicator scale error (4.6.10.1 only)

e. Sealing

f. Response time and warning light

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4.5.2 Sampling tests.

4.5.2.1 Lot. A lot shall consist of 100 systems manufactured under essentially the same conditions and submitted for acceptance at substantially the same time.

4.5.2.2 Sampling plan A. Two complete systems selected at random from each lot shall be subjected to the following tests as described under 4.6:

a. Individual tests

b. System calibration

c. System temperature tests

d. Vibration

e. Detector tests (except burn-in)

4.5.2.3 Sampling plan B. Unless otherwise specified, two complete systems selected at random from the first 10 items of the contract or order shall be subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Humidity

c. Salt fog

d. Electromagnetic interference

e. Indicator adjustment range

f. Indicator lighting

g. Temperature altitude

h. Shock

i. Fire

j. Exposure rate determination

k. Altitude

4.5.2.4 Rejection and retest. When one item from a production lot fails to meet the requirements of this specification; all the instruments still on hand and later produced shall not be accepted by the Government representative until the cause of failure is determined and corrected on all the instruments, at the expense of the contractor. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. An additional sample or sampler from the same lot of instruments shall be selected at random by the Government representative, and shall be subjected to all the inspections specified in 4.4.1. Final acceptance of instruments on hand and later produced shall not be made until the contractor demonstrates to the

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satisfaction of the Government representative that all the instruments meet the requirements of this specification.

4.6 Test methods.

4.6.1 Examination of product. The system shall be examined to determine compliance with the requirements specified herein with respect to materials, weight, finish, marking, workmanship, interchangeability, and dial face markings.

4.6.2 System calibration and voltage and frequency variation test. With the oil tank mounted on a suitable test jig, the system shall be tested throughout +50° to -20° pitch angles in 10° increments and ±15° roll angles in 5° increments. For each combination of pitch and roll, the oil quantity readings shall be determined at each of the following oil tank quantities: 1/16, 1/4, 1/2, 3/4, and full. At each test point, the input power shall be varied in all combinations of 105 to 125V and 360 to 440 Hz. In addition to obtain a test reading at these extremes, a reading shall also be made with 115V, 400 Hz input power. All test readings shall be within the tolerances specified in table I.

4.6.2.1 Adjustment range. The test specified in 4.6.2 shall be repeated except that the tank shall be at 0° pitch and 0° roll only and the source strength shall be reduced to that corresponding to a source which is aged 10 years from the design source strength. Failure to meet the requirements of 3.8.3.3 shall be cause for rejection.

4.6.3 System temperature tests. The source and detector shall be mounted on an oil tank using the bridle assembly and shall be electrically connected to the indicator as shown on figure 2. Two temperature chambers shall be used. The tank and its mounting jig shall be placed in one and the indicators in the other.

4.6.3.1 Low temperature. The temperature in both temperature chambers shall be lowered at a rate not exceeding 10°C per minute to -64°C and maintained at -64° ±2°C (-83 ±2°F) for 48 hours. At the end of this time, the temperature shall be raised from -64° to -54° ±2°C (-65 ±2°F) for the indicators and to -40°C (-4°F) for the source and detector and maintained at these temperatures for 4 hours. At the end of this time, power shall be applied to the master indicator. After 60 ±2 seconds, the first of the readings of oil quantity shall be taken. All readings shall be within ±1/32 of the required quantity. A visual inspection shall also be made for any sign of damage due to low temperature exposure.

4.6.3.2 High temperature. The temperature shall be raised at a rate not exceeding 10° per minute to +93° ±2°C (+167° ±2°F) for the indicators and +204° ±2°C (406° ±2°F) for the source and detector and maintained for 48 hours. The oil may be circulated through the tank and maintained at a temperature of not less than +149°C (+300°F). At the end of the 48-hour period, the temperature shall be reduced on the indicators only to +71° ±2°C (160° ±2°F) and power applied to the indicator and maintained for 4 hours. The relative humidity of the chamber during this test shall not exceed 15 percent. At the end of the 4-hour period, readings of the oil quantity shall be taken. With power on, the temperature shall be allowed to return to room temperature and another set of readings of oil quantity taken. All readings shall be with ±1/32 of the

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required quantity. A visual inspection shall also be made for any sign of damage due to high temperature exposure.

4.6.4 Humidity. The system components, including the bridle, shall be placed in a test chamber and subjected to a humidity test in accordance with MIL-STD-810, Method 507.2, Procedure I. At the end of the test, after the inspection, the source, detector, and bridle shall be installed on the oil tank. Oil quantity readings shall be taken. All readings shall be with ±1/32 of the required reading.

4.6.5 Fungus. The system components, including the bridle assembly, shall be subjected to a fungus test in accordance with MIL-STD-810, Method 508.3. At the end of the test, the source, detector, and bridle shall be installed on the oil tank. Oil quantity readings shall be taken. All readings shall be within ±1/32 of the required reading.

4.6.6 Salt fog. The system components, including the bridle assembly, shall be subjected to a salt fog test in accordance with Method 509.2, Procedure I of MIL-STD-810. At the end of the test, the source, detector, and bridle shall be installed on the oil tank. Oil quantity readings shall be taken. All readings shall be within ±1/32 of the required reading.

4.6.7 Electromagnetic interference. The system, with the source, detector, and bridle installed on the oil tank, shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462 over a frequency range of 0.15 to 400 MHz. Test methods shall be as specified in MIL-STD-461 for class ID equipment.

4.6.8 Vibration. The system components shall be subjected to a vibration test in accordance with MIL-STD-810 and as follows. At the conclusion of this test, the detector, source, and bridle shall be installed on the oil tank. Oil quantity readings shall be taken and shall be within ±1/32 of the required reading. The components shall be so mounted that the specified vibration is transmitted to the equipment under test without attenuation.

a. Indicators: Method 514.3, equipment category (b), Procedure I, part 1, curve C.

b. Detector, source, and bridle: Method 514.3, equipment category (b), Procedure I, part 1, curve L.

4.6.9 Altitude. The system shall be placed in a chamber and mounted on equipment which shall be capable of transmitting vibration at an acceleration of 4 ±0.5g to all components within the chamber without attenuation. The coaxial cable connecting the indicator and detector shall be altered to include three additional coaxial cable connections exclusive of those at the indicator and detector. The connectors used shall be those which are normally supplied with the system as bulkhead, et cetera, connectors. A Kr 85 radioactive source may be substituted for the oil tank and source tube and shall be so positioned that a 1/2-tank indication will be presented by the indicator. Rated power inputs shall be applied, the pressure in the chamber reduced to the equivalent of a 75,000 ft (22860 m) altitude, and the system shall be vibrated at the specified level throughout the test. The system shall be tested at these conditions for a period of 1 hour during which time the indicator shall be observed for any erra-

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tic or unusual indications. Indication changes that deviate from the tolerances specified herein shall constitute failure.

4.6.10 Indicator tests. The following tests shall be conducted on the master and repeater indicators:

4.6.10.1 Indicator scale error. The indicators shall be tested to determine their scale error. The master indicator shall be connected to a master test instrument in lieu of the detector. The master test instrument shall be set to supply simulated detector frequencies corresponding to oil levels of 1/16, 1/4, 1/2, 3/4, and full, and the indicators shall be calibrated. Once set, the adjustments of the master test instrument or the indicators shall not be altered. The indicators shall then be subjected in sequence to the low, room, and high temperature environments. At each temperature, the master test instrument shall supply in sequence the test frequencies and the observed readings shall be recorded. Errors of 1/32 full scale or greater shall be considered a failure.

4.6.10.1.1 Low temperature. The indicators shall be placed in the temperature chamber and subjected to a temperature of -54° ±2°C (-65° ± 2°F) for a minimum period of 4 hours. The master test instrument shall be maintained at room temperature. At the end of this period while the unit is still at low temperature, scale readings shall be determined.

4.6.10.1.2 Room temperature. The indicators shall be held at room temperature (24° ±9°C) for a minimum of 4 hours. At the end of this period, scale readings shall be determined.

4.6.10.1.3 High temperature. The indicators shall be placed in a temperature chamber and subjected to a temperature of 71° ±2°C (160° ±2°F) for a minimum period of 4 hours. The master test instrument shall be maintained at room temperature. At the end of this period and while the unit is still at the high temperature, scale readings shall be determined.

4.6.10.1.4 Room temperature. After the units have returned to room temperature, scale readings shall again be determined with the master test instrument.

4.6.10.2 Indicator cycling. The indicators shall be electrically connected to the master test instrument. During the cycling procedure, the frequency shall be continuously varied in order that the indicator pointers move from zero to end point and return to zero within 5 minutes ±1 minute. A minimum of 5,000 cycles shall be conducted in 12-cycle intervals. After each interval, the indicator shall be de-energized, electrically, a minimum of 10 minutes prior to commencing the next cycle interval. At the conclusion of the cycling test, a scale error test at room temperature only shall be conducted in accordance with 4.6.10.1.

4.6.10.3 Magnetic effect. The indicators, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5 ±1/2 inches (127 ±12.70 mm) from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 225°, and

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315° from the initial position. At each of these positions, the indicator shall be rotated about its own horizontal axis until it is in its normal viewing position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflection of the compass at any of the specified positions shall not exceed 5°.

4.6.10.4 Indicator lighting. The indicator lighting shall be tested in accordance with MIL-L-25467 or MIL-L-27160, as applicable. The dial visibility shall be checked to determine compliance with 3.8.3.6.

4.6.10.5 Sealing. The indicator shall be tested for leaks by means of a mass-spectrometer type of helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per at a pressure differential of approximately 1 atmosphere.

4.6.10.6 Response time and warning light test. The indicator warning light actuation point shall be set at 1/2 full scale and the indicator connected to a suitable lamp. The indicator shall then have a signal input from the master test instrument simulating full scale. An instantaneous change in the signal input to one corresponding to empty shall be applied. The time for the indicator pointer to move to one-half scale shall be measured. The reading at which the light is illuminated shall also be recorded. This procedure shall be repeated with the warning light actuation point set at 1/8 and 3/4 of full scale. The time to reach one-half scale shall be within 8.5 ±0.5 seconds and the warning light actuation point shall be within ±1/32 of the required actuation level.

4.6.10.7 Pointer oscillation. The indicator shall be connected to the detector and the oil level in the tank lowered to approximately one-half full. With a steady state condition achieved, the magnitude of the peak-to-peak oscillation of the point shall be observed. The number of extreme excursions shall also be determined over a 3-minute period. Peak-to-peak oscillation shall not exceed 1/32 of full scale. The frequency of oscillation shall not exceed 1 Hz.

4.6.10.8 Dielectric strength. This test shall be conducted on the indicators prior to hermetic sealing and prior to connection of any internal ground. The specified voltage at a commercial frequency shall be applied between each pin of the connectors and the case. There shall be no breakdown of insulation or any other damage to the indicators as a result of the test. The following test voltages shall be applied for a minimum of 1 minute:

a. Power connector - 500V rms (pin No. 2 of the master indicator is excluded).

b. Coaxial connector - 1.5 times maximum operating voltage (center conductor to case)

4.6.10.9 Thermal shock cycling. The indicator shall be subjected to a total of 8 cycles of immersions in tap water maintained at 85° ±2°C (185° ±2°F) and 5° ±2°C (41° ±2°F). The length of time for each immersion shall be 30 minutes with not more than 3 seconds elapsing between immersions. There shall be no evidence of moisture penetration or damage to the case. At the conclusion of this test, the indicator shall meet the test specified in 4.6.10.5. Fogging of the cover glass or peeling of the paint on the case due to this test shall be cause for rejection.

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4.6.11 Detector tests. For qualification purposes, the detector shall be subjected to the test specified in 4.6.11.2 before the tests specified in 4.6.11.3 through 4.6.11.7 have been conducted and after completion of all these tests and the results recorded. The algebraic sum of the change in the count rate exceeding +7, -3 percent of the initial count rate shall be cause for rejection. The following tests shall be conducted in the sequence specified:

4.6.11.1 Burn-in. The detector shall be placed in a chamber and the temperature maintained at 110° (198°F) for a minimum of 24 hours. Throughout this period, the detector shall be electrically connected as specified in 4.6.11.2 and shall be so positioned with respect to a Kr 85 radioactive source that the master indicator will show approximately a 1/2-tank indication. The test specified in 4.6.11.2 shall be conducted prior to and at the conclusion of this test to determine performance degradation. A change in count rate greater than 1.5 percent of the original count rate shall constitute failure.

4.6.11.2 Detector performance. The detector shall be electrically connected to either a master indication or a pulse-counting circuit with nominal design voltage applied and recorded. A Kr 85 radioactive source shall be so positioned with respect to the detector that the master indicator will show approximately a 1/2-tank indication or an equivalent average count rate will be shown on the counting equipment. This indication, or equivalent count rate, shall be determined as precisely as practicable and shall be recorded. The detector and radioactive source shall be so positioned that the detector may be removed from the setup for the performance of other tests and returned to precisely the same position with the respect of the radioactive source. After removal of the detector for the performance of other specified tests, the detector shall be returned to its former position, connected to the same equipment with the original voltage applied, and the master indicator or count-rate equipment shall be read and recorded.

4.6.11.3 Impact. The test specified in 4.6.11.2 shall be performed before and after this test and the results recorded. A change in count rate of more than ±3 percent shall be considered failure. The detector shall be suspended from a solid pivot point by means of a light flexible cord attached to the electrical connector. The fully extended length of cord shall be such that the distance from the pivot to the geometric center of the detector will be 12 inches (30.48 cm). The pivot point shall be located above the surface of a solid flat, rectangular hardwood table, or equivalent, at a point on a plane which is perpendicular to both the edge and surface of the table. The edge of the table at the point of impact shall be rounded with a radius of 5/16 inch (8 mm). The test shall be conducted as follows:

a. The height of the pivot above the surface of the table shall be adjusted so that with the detector hanging freely, its center of percussion will contact the table. The detector shall then be raised in a pendulous fashion so that the vertical component of the total distance that the center of percussion is displaced will be 2.25 inches (57.15 mm). The detector shall be released and allowed to swing freely to impact with the table edge. This test shall be repeated three times for a total of four impacts.

b. The test specified in subparagraph (a) shall be repeated for an additional four impacts except that the height of the pivot above the table shall be adjusted so that the point of impact is the geometric center of the detector.

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4.6.11.4 Shock. The detector shall be subjected to the test specified in 4.6.11.2 before and after this test and the results recorded. A change in count rate of more than ±3 percent shall be considered failure. The detector shall be dropped freely from the specified height to the surface of a 1-1/2-inch (38.10mm) thick flat hardwood surface. Distances shall be determined by measurements made perpendicular to the surface of the wood to the detector center of gravity. The following three different shock test orientations shall be used, and the drop from each orientation shall be repeated three times:

a. Parallel drop: The detector shall be held parallel to the surface at a height of 16 inches (40.64 cm) and released so that it impacts parallel to the surface.

b. Connector down-angle drop: The detector shall be held with the connector end down 17° ±1° from the horizontal at a height of 10 inches (25.40 cm) from the surface to the center of gravity and released so that the detector impacts the surface at an angle of 17° ±1°, connector end down.

c. Connector up-angle drop: The detector shall be held with the connector end up 17° ±1° from the horizontal at a height of 10 inches (25.40 cm) from the surface to center of gravity and released so that the detector impacts the surface at an angle of 17° ±1°, connector up.

4.6.11.5 Bonding. The detector shall be subjected to the test specified in 4.6.11.2 before and after this test and the results recorded. A change in count rate of more than ± 3 percent shall be considered failure. One end of the detector shall be fastened to a rigid support in such a manner that a bending moment of 20 pound-inches is applied through the centerline of the detector at the unsupported end by means of a jackscrew, or equivalent. While under the specified bending moment, the detector shall be subjected to three cycles of vibration in accordance with MIL-STD-810, Method 514-3, Category 4, Figure 514-3-25, with the direction of vibration perpendicular to the longitudinal axis of the detector. A cycle shall be 20 minutes in length and shall consist of vibrating the equipment as specified for the curves of Fig. 514-3-25.

4.6.11.6 Dielectric strength. A voltage equal to 1.5 times maximum operating voltage rms at a commercial frequency shall be applied between the center conductor and the case. There shall be no breakdown of insulation or other damage to the detector as a result of this test.

4.6.12 Source assembly. The following tests shall be conducted on the source assembly.

4.6.12.1 Shock. Each souce tube supplied per engine installation assembly shall be subjected to a crash safety test in accordance with MIL-STD-810, Method 516.3, Procedure III, Figure 516.3-1, functional test for flight equipment and crash hazard test for flight equipment. The shock shall be 40g peak for a nominal duration of 11 milliseconds. At the conclusion of this test, the source shall be checked with a survey meter to determine that the source tube has not been damaged and that the gas has not escaped.

4.6.12.2 Fire. The source tube shall be brought in direct contact with a flame at every bend, at each end, and with a minimum separation of 4 inches

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(10.16 cm) between flame centers simultaneously a period of 20 minutes. Each flame shall have a minimum temperature of +482°C (900°F) at the point of contact with the tube. At the conclusion of this test, the source shall be checked with the survey meter to determine that no gas has escaped.

4.6.12.3 Exposure rate determination. The source shall be installed on the oil tank. The tank shall be supported at least 4 feet from the ground by a stand which does not extend beyond the outside dimensions of the tank. The stand shall be placed out of doors a minimum of 6 feet from any building or other object. Using the survey meter specified in 4.3.9, exposure rate measurements shall be taken in a horizontal plane 1 foot from the tank at approximately 6-inch intervals around all sides and at each corner of the tank. Exposure rate measurements shall also be taken at a distance of 1 foot in a vertical plane at 6-inch intervals around all sides of the tank and across the tank centerlines. Every reading shall be less than 4 milliroentgens per hour.

4.6.12.4 Contact exposure rate. Using the survey meter specified in 4.3.9, the contact exposure rate shall be measured. The survey meter face shall be placed in direct contact with the source tube along the diameter of the face. Measurements shall be made, with and without the beta shield installed, at a minimum of three locations along the length of tube.

4.6.13 Reliability and longevity

4.6.13.1 Reliability test program. Prior to conducting the reliability test, the contractor shall prepare a reliability test program which shall be approved by the procuring activity.

4.6.13.2 System reliability. Five systems shall be subjected to a reliability test in accordance with MIL-STD-781, test level E, except the temperature cycling shall be between -40° and +71°C (-40° and +160°F). Accept-reject criteria shall be in accordance with test plan I. Recording, data handling, and reporting shall also be in accordance with MIL-STD-781.

4.6.13.3 Detector reliability. Eight detectors shall be subjected to a reliability test in accordance with MIL-STD-781, test level E, except that the temperature cycling shall be between -40°C and +175°C (-40°F and +315°F) and the acceleration of the vibration shall be 4 $\pm$0.2g. The detectors shall have nominal system operation voltage applied during operation ON periods and shall be in the presence of a radioactive source of such strength and energy that a 3/4-tank indication is presented if the detectors are individually coupled to an indicator. Nominal system operating voltage shall be applied to the detectors, shall be recorded, and shall not be varied throughout the test. Each detector shall be subjected to the test specified in 4.6.11.2 before and after this test and the results recorded. A change in average count rate which exceeds a tolerance of +7° -3° of the initial count rate shall be considered a failure. Accept-reject criteria shall be in accordance with test plan I of MIL-STD-781.

4.6.13.4 Detector output. A minimum of ten 10-second counts shall be used to determine the average count rate reported by the detector.

4.6.13.5 Longevity. Testing of two of the five systems specified in 4.6.13.2 shall be continued through a longevity test in accordance with MIL-STD-781, test plan XXVIII.

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4.6.14 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471, appendix B, method 1.

* 5. PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be level A & C or Commercial/Industrial as specified (See 6.2).

* 5.1.1 Level A. The components of oil quantity gaging system shall be packaged in accordance with method III of MIL-P-116, except the system indicator shall be packaged in accordance with method IA.

* 5.1.2 Level C. The oil quantity gaging system shall be provided with sufficient protection against physical and mechanical damage to assure safe delivery without degradation of the item reliability from the supply source to the first receiving activity for immediate use.

5.1.3 Commercial. Commercial/industrial preservation of oil quantity gaging system shall be in accordance with the requirements of ASTM D-3951.

* 5.2 Packing.

* 5.2.1 Packing. Packing shall be level A, B, C or commercial/industrial, as specified (See 6.2).

* 5.2.2 Level A. The oil quantity gaging system preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-601, overseas type. The closure shall be in accordance with the appendix of the container specification.

* 5.2.3 Level B. The oil quantity gaging system preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-636, weather resistant. The closure shall be in accordance with the appendix of the container specification.

* 5.2.4 Level C. The oil quantity gaging systems preserved and packaged as specified in 5.1.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Container shall be in accordance with either the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.2.5 Commercial. Commercial/industrial packing of oil quantity system shall be in accordance with the requirements of ASTM D-3951.

* 5.3 Physical protection. Cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cushioning materials and container shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged, and packed for immediate use of when the drop tests of MIL-P-116 are applicable.

* 5.4 Marking. In addition to any other marking required by the order or contract, interior and exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The nucleonic oil quantity gaging systems covered by this specification are intended for use in aircraft to indicate the quantity of oil contained in the engine oil tank.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article inspections (4.4).

c. When a repeater indicator is required. When possible, it should specify the number of radiation sources to be furnished with each system.

d. Levels of packaging and packing required (See section 5).

* 6.3 Data Required. For the purpose of first article tests the manufacturer will furnish three copies of data containing the following information:

a. Gaging system part number

b. Wiring diagram

c. List of all components with the manufacturer's part number

d. Data for sources

(1) Total strength in millicuries

(2) Maximum dosage at 1 foot from the tank per hour

(3) Type of sources and method of shielding, if any

(4) Drawing of installation

e. Data for detector

(1) Maximum operating voltage

(2) Drawing of installations

f. Drawing of indicator

g. Kind and type of cabling and wiring

h. Reliability analysis.

6.4 Definition

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6.4.1 Hermetic seal. A hermetic seal is a perfectly closed and airtight seal made between vitric or metallic materials. A hermetic seal does not include seals accomplished by gasket or O-rings.

* 6.5 First Article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The test samples shall consist of seven complete systems including repeater indicators. Five systems shall be subjected to the reliability tests only and the other two systems shall be subjected to all tests except reliability. The test samples shall be identified with the manufacturer's own part numbers. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.6 Subject term (keyword) listing.

Oil quantity gaging system, continuous transistorized, nucleonic

Oil quantity gaging system

Gaging system, oil quantity

Continuous transistorized, nucleonic, oil quantity gaging system

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99
Army - AV

Review activity:
DLA-GS

Preparing activity:
Air Force - 82

Project No. 6680-0192

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